04/11/2026

Code-C_Phase-3

Aakash
Medical | IIT-JEE | Foundations

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Fortnightly Test for NEET-2027 (XII-Passed)_FT04C

Time : 180 Min.

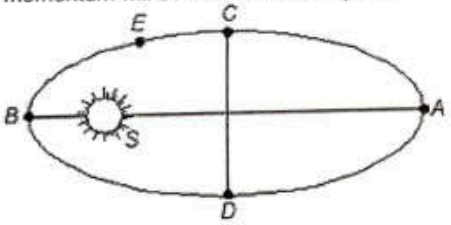
MM : 720

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry: Equilibrium, Redox Reactions, Electrochemistry
Botany: Anatomy of Flowering Plants, Plant Kingdom
Zoology: Excretory Products & their Elimination, Locomotion & Movement

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. The earth *E* moves in an elliptical orbit with the sun *S* at one of the focii as shown in figure. The magnitude of its linear momentum will be minimum at the point



(1) C (2) B
(3) A (4) D

2. Two planets *A* and *B* have same density. The radius of *A* is twice that of *B*. The ratio of escape speeds from planet *A* to the planet *B* $V_A : V_B$ is

(1) 2 : 1 (2) $\sqrt{2}$: 1
(3) 1 : $\sqrt{2}$ (4) 1 : 2

3. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius *R* is proportional to:

(1) $R^{3/2}$ (2) R^2
(3) R^3 (4) $R^{1/2}$

4. If a body of mass *m* is released from a height 2*R* from the Earth's surface (where *R* is radius of earth), then the kinetic energy of body when it reaches at a height *R* from the earth's surface is (*M* is mass of earth)

(1) $\frac{2GMm}{3R}$ (2) $\frac{GMm}{2R}$
(3) $\frac{3GMm}{2R}$ (4) $\frac{GMm}{6R}$

5. If an object weights 50 N on the surface of earth, then mass of the object under free fall is

(1) 50 kg (2) zero
(3) 5 kg (4) 10 kg

6. A : Total energy of a satellite in an elliptical orbit remains constant.
R : The power of gravitational force on satellite is always zero in the elliptical path.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

7. The height above the earth's surface at which the weight of a person becomes $\frac{1}{4}^{th}$ of his weight on the surface of earth is (*R* is the radius of earth)

(1) 2 *R* (2) *R*
(3) 3 *R* (4) 5 *R*

8. Two particles of masses m_1 and m_2 are placed at a distance *r*. The ratio of gravitational force on mass m_1 due to mass m_2 and on mass m_2 due to mass m_1 is

(1) $m_1 : m_2$ (2) $m_1^2 : m_2^2$
(3) 1 : 1 (4) $\sqrt{\frac{m_1}{m_2}}$

9. An astronaut in a strange planet observes that a maximum horizontal distance of 4 m, can be jumped if the initial speed is 8 m s^{-1}. What is the acceleration due to gravity of the planet?

(1) 8 m s^{-2} (2) 10 m s^{-2}

(3) 16 m s^{-2} (4) 64 m s^{-2}

10. Consider the statements and choose the correct option.

Statement A: A uniform spherical shell of matter attracts a particle that is outside the shell as if all of its mass were concentrated at its centre.

Statement B: The reading of a barometer on moon is zero.

(1) Statement A is correct but statement B is incorrect

(2) Statement B is correct but statement A is incorrect

(3) Both statements are correct

(4) Both statements are incorrect

11. A body of mass m is projected horizontally just above the Earth's surface with a velocity $\sqrt{\frac{23}{12}gR}$, where g is acceleration due to gravity at Earth's surface and R is radius of earth. The trajectory of body will be

(1) Hyperbola (2) Circle

(3) Ellipse (4) Parabola

12. The weight of a person is W at the surface of earth. At what height from surface of earth the weight reduces by 36%? (R is radius of earth)

(1) R (2) $\frac{R}{2}$

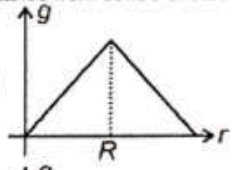
(3) $\frac{R}{4}$ (4) $\frac{R}{3}$

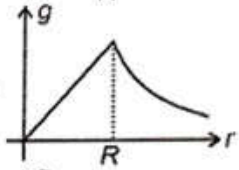
13. According to Kepler's 3rd law $T^2 = Ka^3$, where K is proportionality constant which depends on

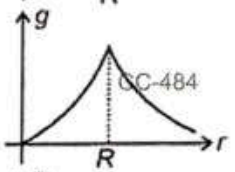
(1) Mass of sun (2) Mass of planet

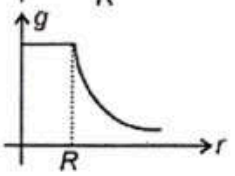
(3) Radius of sun (4) Both (1) & (2)

14. Which graph among the given options shows the most appropriate variation of acceleration due to gravity (g) with distance from centre of earth (where R is radius of earth)

(1)


(2)


(3)


(4)


15. If the speed of an artificial satellite revolving around the earth in circular orbit be $\frac{2}{3}$ of escape velocity from the surface of the earth then its altitude above the surface of earth is

(1) $\frac{4}{5}R$ (2) $\frac{2}{5}R$

(3) $\frac{1}{8}R$ (4) $\frac{3}{5}R$

16. If the earth suddenly stops rotating, then the value of acceleration due to gravity g at equator and pole respectively

(1) Decreases, decreases

(2) Increases, remains constant

(3) Decreases, remains constant

(4) Remains constant, decreases

17. A metallic wire is fixed at one end of roof. At other end of the wire, a 2 kg mass is hung so that energy stored in the wire is U. The energy stored in the same wire if a mass of 4 kg is hung instead of 2 kg mass, will be

(1) U (2) $2U$

(3) $3U$ (4) $4U$

18. The breaking stress of a metal is 6.4×10^5 N/m^2. The maximum length of the metallic wire having density 3.2×10^3 kg/m^3 that can hang vertically without getting broken is

(1) 2 m (2) 0.2 m

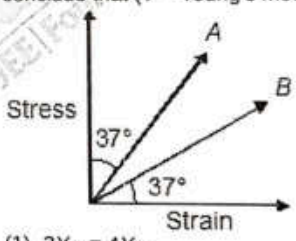
(3) 4 m (4) 20 m

19. Copper of fixed volume V is drawn into wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. Which of the following graphs is a straight line?

(1) Δl versus l^2 (2) Δl versus l

(3) Δl versus $\frac{1}{l}$ (4) Δl versus $\frac{1}{l^2}$

20. From given stress-strain curve for two materials, we conclude that (Y = Young's modulus)



(1) $3Y_B = 4Y_A$ (2) $9Y_A = 16Y_B$

(3) $4Y_B = 3Y_A$ (4) $9Y_B = 16Y_A$

21. The bulk modulus of water is 3×10^9 N/m^2. The extra pressure required to increase the density of water by 0.2% is

(1) 1.5×10^6 N/m^2 (2) 3×10^6 N/m^2

(3) 6×10^6 N/m^2 (4) $\frac{2}{3} \times 10^6$ N/m^2

22. A rubber ball is taken to depth 2 km inside water so that its volume reduces by 0.04%. The bulk modulus of rubber is (in N/m^2)

(1) 2×10^8 (2) 5×10^{10}

(3) 1×10^{10} (4) 4×10^8

23. Which of the following statements is wrong?

(1) Young's modulus for rubber is zero

(2) Bulk modulus is relevant for solids, liquids and gases

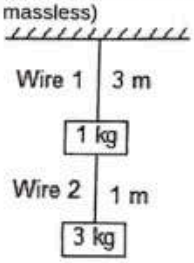
(3) Young's modulus for perfectly rigid body is infinity

(4) Rubber is less elastic than steel

24. Elasticity of a substance cannot be affected by
(1) Change in temperature
(2) Hammering and annealing
(3) Impurity in substance
(4) Cutting the substance into pieces

25. Tensile stress is possible in
(1) Liquids (2) Gases
(3) Solids (4) Both (1) & (3)

26. Two blocks of masses 1 kg and 3 kg are suspended with the help of two wires having same area of cross section. If the ratio of Young's modulus of wire 1 and wire 2 is 1 : 3, then the ratio of extension produced in wires is (Assume wire is massless)



(1) 1 : 3 (2) 3 : 1
(3) 1 : 12 (4) 12 : 1

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27. The bulk modulus of a spherical object is B. If it is subjected to uniform pressure P, the fractional decrease in radius is ($P << B$)
(1) $\frac{P}{B}$ (2) $\frac{B}{3P}$
(3) $\frac{P}{3B}$ (4) $\frac{B}{P}$

28. A metal rope of density 5000 kg/m^3 has a breaking stress 9.8×10^8 N/m^2. If the radius of the rope is doubled, then the new breaking stress will be
(1) 39.2×10^8 N/m^2 (2) 9.8×10^8 N/m^2
(3) 6×10^8 N/m^2 (4) 16×10^8 N/m^2

29. Hooke's Law is obeyed within
(1) Elastic limit (2) Proportionality limit
(3) Breaking point (4) Freezing point

30. One end of a uniform wire of length L and of mass M is attached rigidly to a point on the ceiling and a block of mass M is suspended from its lower end. If area of cross-section of the wire is A, then stress in the wire at a distance $\frac{3L}{4}$ from its lower end will be
(1) $\frac{Mg}{4}$ (2) $\frac{3Mg}{4A}$
(3) $\frac{2Mg}{A}$ (4) $\frac{7Mg}{4A}$

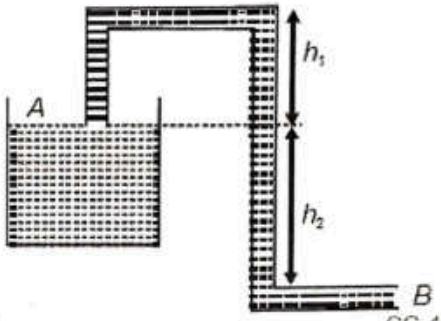
31. Pressure at the bottom of a tank of water is $3P$, where P is atmospheric pressure. If water is drawn out till the level of water is lowered by two-fifth, then the pressure at the bottom of the tank will be
(1) $\frac{6P}{5}$ (2) $\frac{13P}{5}$
(3) $\frac{9P}{5}$ (4) $\frac{11P}{5}$

32. A solid block of density 480 kg/m^3, floats in a liquid of density 800 kg/m^3. The percent of volume which is outside the liquid is
(1) 20% (2) 40%
(3) 50% (4) 30%

33. A sphere of mass M and radius R is falling in a viscous liquid. The terminal velocity attained by the sphere will be proportional to
(1) R^2 (2) R
(3) $\frac{1}{R}$ (4) $\frac{1}{R^2}$

34. Bernoulli's equation is consequence of conservation of
(1) Mass (2) Linear momentum
(3) Energy (4) Angular momentum

35. From given diagram the speed with which water leaves the tube B of small diameter is



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(1) $\sqrt{2gh_1}$ (2) $\sqrt{2gh_2}$
(3) $\sqrt{2g(h_1+h_2)}$ (4) $7\sqrt{2g(h_2-h_1)}$

36. The surface tension of a liquid is 5 Nm^{-1}. If a film is held on a ring of area 0.02 m^2, then its total surface energy is about
(1) 3×10^{-2} J (2) 2×10^{-1} J
(3) 3×10^{-3} J (4) 10^{-1} J

37. The property of surface tension of liquid is due to
(1) Gravitational force of attraction between molecules
(2) Cohesive forces between molecules
(3) Adhesive force between molecules
(4) Formation of ionic bonds among molecules

38. A liquid filled in a container has plane meniscus. If θ is angle of contact, then
(1) $\theta = 0^\circ$ (2) $\theta = 90^\circ$
(3) $\theta < 90^\circ$ (4) $\theta = 180^\circ$

39. Eight droplets of water each of radius 0.5 mm combine to form a bigger drop. The energy evolved in this process is (surface tension = 7.2×10^{-2} N/m)
(1) 9×10^{-7} J (2) 3×10^{-7} J
(3) 1×10^{-7} J (4) 10×10^{-7} J

40. An orifice of radius r is present at the bottom of a container. The maximum height to which the container can be filled with a liquid of density ρ for which liquid will not come out of the orifice is

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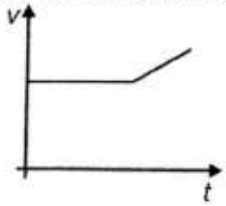
(1) $\frac{T}{r\rho g}$ (2) $\frac{3T}{r\rho g}$
(3) $\frac{2T}{r\rho g}$ (4) $\frac{4T}{r\rho g}$

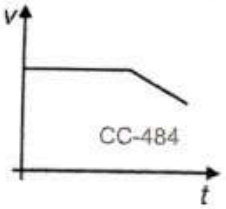
41. A large open tank with a square hole of side 0.1 cm in the wall at a depth of 0.2 m from the top is completely filled with a liquid. The rate of flow of liquid (in cm^3/s) through the hole will be
(1) 1 (2) 2
(3) 3 (4) 4

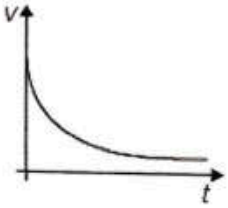
42. The diameter of a syringe is 4 mm and the diameter of its nozzle (opening) is 1 mm. The syringe is placed on the table horizontally at a height of 1.25 m. If the piston is moved at a speed of 0.5 m/s, then considering the liquid in syringe to be ideal, the horizontal range of liquid is ($g = 10\ m/s^2$)

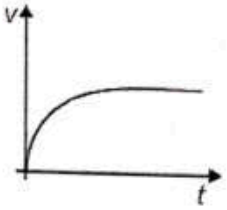
(1) 4 m (2) 8 m
(3) 0.4 m (4) 0.2 m

43. For amongst the following curves which one shows the variation of velocity with time (t) for small sized spherical body falling vertically in a long column of viscous liquids

(1) 

(2) 

(3) 

(4) 

44. Match the **column I** with **column II** and select the correct option regarding shape of meniscus in capillary rise of liquid.

	Column I (Angle of contact)		Column II (Liquid)
a.	$\theta < 90°$	(i)	Concave down Meniscus
b.	$\theta = 90°$	(ii)	Concave up Meniscus
c.	$\theta > 90°$	(iii)	Capillary rise is zero

(1) a(ii), b(iii), c(i) (2) a(iii), b(ii), c(i)
(3) a(i), b(ii), c(iii) (4) a(i), b(iii), c(ii)

45. Consider the following statements and identify the correct answer.

Statement (A): The breaking stress of a wire depends on nature of material of the wire.

Statement (B): For an ideal liquid, the shear modulus is infinite and the bulk modulus is zero.

Statement (C): Shear modulus (or modulus of rigidity) is generally less than Young's modulus for most common isotropic materials.

(1) Only A is true (2) A and B both are true
(3) A and C both are true (4) A, B and C all are true

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CHEMISTRY

46. Molar solubility of $M(OH)_2$ in 0.20 M NaOH is (Given K_{sp} of $M(OH)_2 = 4.0 \times 10^{-15}$)

(1) 10^{-13} M (2) 10^{-15} M
(3) 10^{-11} M (4) 10^{-17} M

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47. When 50 mL of 0.1 M NH_4OH is mixed with 25 mL of 0.1 M HCl, then pH of the resulting mixture is [$pK_b = 4.74$]

(1) 4.74 (2) 9.26
(3) 7 (4) 10

48. Two solids dissociate in a container as follows:

$A(s) \rightleftharpoons B(g) + C(g)$; $K_{p_1} = a\ atm^2$

$D(s) \rightleftharpoons C(g) + E(g)$; $K_{p_2} = b\ atm^2$

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The total equilibrium pressure when both the solids dissociate simultaneously is:

(1) (a + b) atm (2) $2\left(\sqrt{a+b}\right)$ atm
(3) $(a + b)^2$ atm (4) $\frac{a+b}{\sqrt{2}}$ atm

49. Given below are two statements: one is labelled as **Assertion (A)** the other is labelled as **Reason (R).**
Assertion (A): Upon addition of $HgCl_2(aq)$ to the following reaction mixture:
$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$;
The intensity of red colour decreases.
Reason (R) : Hg^{2+} reacts with SCN^- to form stable complex ion $[Hg(SCN)_4]^{2-}$.
In the light of above statements, select the most appropriate option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

50. Match the List I and List II.

	List I (Salt)		List II (solubility in terms of solubility product K_{sp})
a.	$BaSO_4$	(i)	$\left(\frac{K_{sp}}{108}\right)^{1/5}$
b.	Hg_2Cl_2	(ii)	$\left(\frac{K_{sp}}{27}\right)^{1/4}$
c.	$Ca_3(PO_4)_2$	(iii)	$\left(\frac{K_{sp}}{4}\right)^{1/3}$
d.	$Fe(OH)_3$	(iv)	$(K_{sp})^{1/2}$

The correct match is
(1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(i), c(iv), d(ii)

51. Given below are two statements
Statement (I) : If equilibrium constant is 10^{-12} then product will be present predominantly at equilibrium state.
Statement (II) : If molar concentrations of products and reactants are same at equilibrium then value of K_c will always be 1.
In the light of above statements, choose the **correct** option.
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

52. Consider the following statements about catalyst
(a) It lowers the activation energy for the forward and reverse reactions by exactly the same amount
(b) It increases the equilibrium constant value of the reaction
(c) it does not appear in the equilibrium constant expression
The correct statements are
(1) (a) and (b) only (2) (a) and (c) only
(3) (a), (b) and (c) (4) (b) and (c) only

53. If at equilibrium HI dissociates 50% at 25°C, then ΔG° for the reaction, $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ is
(1) 1.386 RT (2) –2.303 RT
(3) 0.693 RT (4) –0.693 RT

54. Which pair of solution can form an acidic buffer?
(1) H_2CO_3 and $NaHCO_3$ (2) H_2CO_3 and H_2SO_4
(3) HNO_3 and CH_3COOH (4) NH_4Cl and HCl

55. The K_a value of HNO_2 at 25°C is 4×10^{-4}. The pK_b of NO_2^- ion at 25°C is (log 2.5 = 0.4)
(1) 10.6 (2) 2.5×10^{-11}
(3) 2.5×10^{11} (4) 3.4

56. For the reaction
$FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$
If the initial partial pressure of CO is 4 atm, then the equilibrium partial pressure of CO will be (K_p = 1.5)
(1) 2.4 atm (2) 1.6 atm
(3) 3 atm (4) 5.5 atm

57. If K_1 and K_2 are respective equilibrium constants for the two reactions.
$XeF_6(g) + H_2O(g) \rightleftharpoons XeOF_4(g) + 2HF(g)$; (K_1)
$XeO_4(g) + XeF_6(g) \rightleftharpoons XeOF_4(g) + XeO_3F_2(g)$; (K_2)
The equilibrium constant of the reaction
$XeO_4(g) + 2HF(g) \rightleftharpoons XeO_3F_2(g) + H_2O(g)$ will be
(1) $\frac{K_1}{(K_2)^2}$ (2) $K_1 \times K_2$
(3) $\frac{K_1}{K_2}$ (4) $\frac{K_2}{K_1}$

58. Match reaction in List-I with their correct relation of K_p and K_c in List-II

List-I	List-II
(a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(i) $K_p = K_c(RT)^{-2}$
(b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(ii) $K_p = K_c(RT)$
(c) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(iii) $K_p = K_c$
(d) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(iv) $K_p = K_c(RT)^{-1}$

Choose the **correct** match
(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

59. Which of the following compound behave as a Lewis acid?
(1) $AlCl_3$ (2) H_2O
(3) CN^- (4) NH_3

60. If 100 mL solution of pH = 2 is mixed with 100 mL of solution of pH = 3, then pH of mixture will be
(1) 1.5 (2) 2.26
(3) 0.39 (4) 3.5

61. The pH of 0.1 M sodium acetate solution is [Given pK_a of acetic acid = 4.76]
(1) 9.8 (2) 8.9
(3) 5.1 (4) 9.4

62. The equilibrium constant of the reaction can be altered by
(1) Adding reactants (2) Adding products
(3) Increasing volume (4) Decreasing temperature

63. Consider the following reaction
$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g)$, $\Delta H = -92.2$ kJ
Choose the condition(s) from the following which favour(s) the formation of methanol.
(a) Decreasing the pressure.
(b) Addition of inert gas at constant volume
(c) Decreasing the temperature
(1) (a) only (2) (a) and (b) only
(3) (c) only (4) (a) and (c) only

64. The pH of 10^{-8} M NaOH solution is
(1) 6.98 (2) 6
(3) 7.02 (4) 8

65. Consider the following statements
(a) Cl exhibits its highest oxidation state in $NaClO_3$.
(b) Conversion of $Cr_2O_7^{2-}$ to CrO_4^{2-} does not require a reducing agent.
(c) Cl exists in both positive and negative oxidation state in $CaOCl_2$.
Correct statement(s) is/are
(1) (c) only (2) (a), (b) and (c)
(3) (b) and (c) only (4) (a) only

66. Consider the given redox reaction in acidic medium
$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$
The number of moles of $KMnO_4$ required to oxidise 0.5 mol of oxalate ions is
(1) 0.1 mol (2) 0.2 mol
(3) 0.5 mol (4) 1 mol

67. Given below are two statements
Statement I: Oxidation state of terminal Br in Br_3O_8 is +7.
Statement II: Average oxidation number of Br in Br_3O_8 is $\frac{+16}{3}$.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

68. If oxidation numbers of A, B, C are +1, +4, –3 respectively then possible formula of compound will be
(1) ABC (2) A_3BC_2
(3) ABC_3 (4) A_2BC_2

69. Match reactions in List-I with their type in List-II.

List-I	List-II
(a) $KClO_3 \xrightarrow{\Delta} KCl + O_2$	(i) Non-redox
(b) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	(ii) Non-metal displacement redox
(c) $Na + H_2O \rightarrow NaOH + H_2$	(iii) Metal displacement redox
(d) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$	(iv) Decomposition redox

Correct match is
(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)
(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)
(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

70. Oxidation state of Cr in CrO_5 is
(1) +10 (2) +8
(3) +6 (4) +4

71. Oxidation number of Fe in Fe_3O_4 is
(1) + 2 only (2) + 3 only
(3) + 2 and + 3 (4) + 4

72. The standard reduction potential of Mg, Al, Zn and Na are respectively –2.36 V, –1.66 V, –0.76 V and –2.71 V. The correct decreasing order of reducing power of metal is
(1) Mg > Na > Al > Zn (2) Zn > Mg > Na > Al
(3) Na > Mg > Al > Zn (4) Zn > Al > Mg > Na

73. Given below are two statements, one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): E_{cell} is an intensive property.
Reason (R): As per the equation, $\Delta G = -nFE_{cell}$, the value of E_{cell} depends on n.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

74. If for the electrochemical cell
A(s) | A^{2+} (0.1 M) || B^+(0.1 M) | B(s) the emf is 5 V, then E°_{cell} of the reaction is
(1) 6.78 V (2) 5.03 V
(3) 2.18 V (4) 10.07 V

75. Amount of Al deposited on passage of 2 F charge through molten Al_2O_3 is (Atomic mass of Al = 27 u)
(1) 9 g (2) 15 g
(3) 18 g (4) 27 g

76. E°_{cell} of cell $Pb|Pb^{2+}||Hg^{2+}, Hg_2^{2+}|Pt$, is
$\left(\text{given}: E^\circ_{Pb^{2+}/Pb} = -0.13\ V \text{ and } E^\circ_{Hg^{2+}/Hg_2^{2+}} = 0.97\ V\right)$
(1) 1.2 V (2) 1.1 V
(3) 1.3 V (4) 0.84 V

77. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 5 and by passing hydrogen gas around the platinum wire at one atm pressure at 298 K temperature. The reduction potential of the electrode would be
(1) –0.295 V (2) –0.148 V
(3) +0.295 V (4) +0.148 V

78. Unit of cell constant is
(1) cm^{-1} (2) Ohm^{-1}
(3) Ohm cm^{-1} (4) Ohm cm

79. Match batteries in list-I with their electrolyte in List-II.

	List-I		List-II
(a)	Dry cell	(i)	38% aq H_2SO_4
(b)	Mercury cell	(ii)	Paste of NH_4Cl and $ZnCl_2$
(c)	$(H_2–O_2)$ Fuel cell	(iii)	Paste of KOH and ZnO
(d)	Lead storage battery	(iv)	Conc. (aq) NaOH

Correct match is
(1) (a) (ii), (b) (iii), (c)(i), (d)(iv)
(2) (a) (iii), (b) (ii), (c)(i), (d) (iv)
(3) (a)(iii), (b) (ii), (c) (iv), (d) (i)
(4) (a) (ii), (b) (iii), (c) (iv), (d) (i)

80. The quantity of electricity required to convert 0.5 mol of MnO_4^- into Mn^{2+} is
(1) 5 × 96500 C (2) 2.5 × 96500 C
(3) 1 × 96500 C (4) 0.5 × 96500 C

81. Given : (i) $Cu^+ + e^- \rightarrow Cu$, $E^\circ = 0.52$ V
(ii) $Cu^{2+} + 2e^- \rightarrow Cu$, $E^\circ = 0.34$ V
Standard reduction potential, E° for the reaction, $Cu^{2+} + e^- \rightarrow Cu^+$ will be
(1) 0.16 V (2) 0.30 V
(3) 0.18 V (4) 0. 86 V

82. Identify the correct anode and cathode respectively in mercury cell.
(1) Paste of KOH + ZnO and paste of HgO + C
(2) Zn-Hg amalgam and paste of KOH + ZnO
(3) Zn-Hg amalgam and paste of HgO + C
(4) Paste of KOH + C and Zn-Hg amalgam

83. Consider the following statements:
Statement I: During corrosion, at anode, Fe gets oxidised to Fe^{2+} ion.
Statement II: During corrosion, at cathode, $O_2(g)$ gets reduced in presence of H^+ to form H_2O.
In the light of above statements, choose the correct option.
(1) Both statement I and statement II are correct
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and statement II are incorrect

84. Product of electrolysis of aqueous NaCl solution using Pt electrode are
(1) $Cl_2(g)$ and $H_2(g)$ (2) $O_2(g)$ and $H_2(g)$
(3) Na(s) and $Cl_2(g)$ (4) $Cl_2(g)$ and $O_2(g)$

85. If conductivity of 0.2 M solution of KCl at 300 K is 0.015 S cm^{-1} then its molar conductivity (in S cm^2 mol^{-1}) is
(1) 75 (2) 50
(3) 7.5 (4) 5

86. The molar conductivity of M/10 solution of a weak monobasic acid is 20 S cm^2 mol^{-1}. If the molar conductivity at infinite dilution is 600 S cm^2 mol^{-1}, then dissociation constant of the acid is nearly
(1) 1×10^{-4} (2) 2×10^{-4}
(3) 3×10^{-5} (4) 2×10^{-5}

87. On passing same quantity of electricity through cells connected in series containing solutions of $AgNO_3$ and $CuSO_4$, 0.54 g of silver is deposited. The amount of copper deposited will be
(1) 0.158 g (2) 0.3175 g
(3) 0.079 g (4) 0.395 g

88. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and 91.0 S cm^2 mol^{-1} respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.
(1) 540.48 S cm^2 mol^{-1} (2) 201.28 S cm^2 mol^{-1}
(3) 390.71 S cm^2 mol^{-1} (4) 698.28 S cm^2 mol^{-1}

89. Select the correct statement among the following
(1) Molar conductivity of strong electrolyte increases with increase of dilution
(2) Conductivity of weak electrolyte decreases on increase in concentration
(3) Conductivity of strong electrolyte increases with increase of dilution
(4) Molar conductivity increases to two times when cell constant becomes double

90. Limiting molar conductivity of which ion is highest at 298 K in water?
(1) H^+ (2) Na^+
(3) Mg^{2+} (4) Ca^{2+}

BOTANY

91. The trichomes in the shoot system
(1) Are usually multicellular
(2) Can be branched or unbranched but never stiff
(3) Cannot be secretory in function
(4) Cannot help in preventing water loss due to transpiration

92. Guard cells are bordered by subsidiary cells. Subsidiary cells are
(1) Highly lignified epidermal cells
(2) Modified epidermal cells
(3) Rich in chloroplast
(4) Highly suberised cells

93. Unicellular root hairs arise from
(1) Epiblema (2) Endodermis
(3) Pericycle (4) Pith

94. Bundle sheath and conjoint-closed vascular bundles are present in all given parts of plants, **except**
(1) Stem of dicotyledonous plants
(2) Stem of grasses
(3) Dorsiventral leaves
(4) Isobilateral leaves

95. All tissues given below constitute the ground tissue system, **except**
(1) Endodermis (2) Pericycle
(3) Hypodermis (4) Vascular bundles

96. Stomatal apparatus does **not** include
(1) Stomatal pore
(2) Guard cells
(3) Surrounding subsidiary cells
(4) Hypodermal cells

97. In dicot stem, hypodermis is made up of
(1) Parenchyma (2) Collenchyma
(3) Sclerenchyma (4) Sclereids

98. Select the **odd** one out w.r.t. T.S. of the dicot root.
(1) Cambium develops at the time of secondary growth.
(2) There are usually more than six xylem bundles.
(3) Cells of the cortex are loosely arranged which allows the free movement of water through them.
(4) Presence of conjunctive tissue.

99. Closed vascular bundle differs from open vascular bundle as in the latter
(1) Xylem and phloem are jointly situated on the same radius.
(2) Xylem and phloem are separated by collenchymatous cells.
(3) Cambium is present between xylem and phloem.
(4) Xylem is not found on the same radii on which phloem is present.

100. In the dicot roots the lateral roots originates from
(1) Parenchyma between endodermis and pericycle
(2) Innermost region of the stele
(3) Parenchymatous cells present next to endodermis
(4) Intrafascicular and interfascicular tissue in a ring

101. Cortex is the region found between
(1) Epidermis and stele
(2) Pericycle and endodermis
(3) Endodermis and pith
(4) Endodermis and vascular bundle

102. Select the **incorrect** option w.r.t. conjunctive tissue.
(1) It is made up of parenchymatous cells
(2) It is present in between the patches of xylem and phloem
(3) It is found only in monocotyledonous roots
(4) It is a part of stele

103. Read the given statements and choose the **correct** option.
Statement (A) : In stems of dicotyledonous plants, all the three subzones of cortex consist of collenchymatous cells which provide strength to them.
Statement (B) : The presence of casparian strips in the outermost layer of stele region is characteristic feature of the angiospermic roots.
(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both statements (A) and (B) are incorrect
(4) Both statements (A) and (B) are correct

104. Starch sheath is present in
(1) Dicot root (2) Dicot stem
(3) Monocot root (4) Monocot stem

105. Water-containing cavities are present within the vascular bundles of
(1) Dicot stem (2) Monocot stem
(3) Dicot root (4) Monocot root

106. Hypodermis is
(1) Innermost layer of cortex
(2) Outermost sub-zone of cortex in stems
(3) Multilayered in roots
(4) Collenchymatous in monocots

107. Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are
(1) Guard cells (2) Lenticels
(3) Bundle sheath cells (4) Bulliform cells

108. Select the odd one w.r.t dorsiventral leaf.
(1) Reticulate venation
(2) Large colourless cell on adaxial epidermis
(3) Mesophyll has two type of cells
(4) Abaxial epidermis with more stomata than adaxial epidermis

109. A large number of scattered vascular bundles, each surrounded by sclerenchymatous bundle sheath can be observed in
(1) Monocot root (2) Dicot stem
(3) Monocot stem (4) Dicot root

110. Mesophyll tissue is differentiated into palisade and spongy parenchyma in
(1) Grass leaf (2) Monocot leaf
(3) Isobilateral leaf (4) Dicot leaf

111. The most appropriate reason for curling of grass leaves inwards during water stress condition is
(1) Formation of spongy parenchyma
(2) Flaccidity of bulliform cells
(3) Closure of stomata
(4) Due to presence of air cavity

112. Classification for flowering plants given by George Bentham and Joseph Dalton Hooker
(1) Is also known as artificial system of classification
(2) Based mainly on the androecium structure
(3) Considers embryology, ultrastructure, phytochemistry
(4) Uses only gross superficial morphological characters

113. Cytotaxonomy is based on all of the following, **except**.
(1) Chromosome number (2) DNA sequencing
(3) Chromosome structure (4) Chromosome behaviour

114. Algin and carrageen are respectively produced by the members of
(1) Rhodophyceae and phaeophyceae
(2) Phaeophyceae and chlorophyceae
(3) Chlorophyceae and phaeophyceae
(4) Phaeophyceae and rhodophyceae

115. Match the columns and choose the **correct** option.

	Column-I		**Column-II**
I.	Syngamy	a.	Morphologically distinct gametes
II.	Oogamy	b.	Union of two gametes
III.	Anisogamy	c.	Two kinds of gametes, one motile male and other non-motile female
IV.	Isogamy	d.	Morphologically indistinguishable gametes

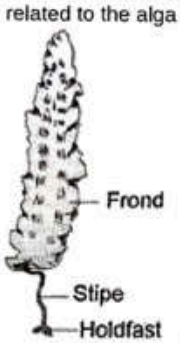
(1) I-(b), II-(c), III-(d), IV-(a) (2) I-(b), II-(c), III-(a), IV-(d)
(3) I-(c), II-(a), III-(d), IV-(b) (4) I-(c), II-(b), III-(a), IV-(d)

116. Read the following statements and select the **correct** option.
Assertion (A) : Being photosynthetic, algae increase the level of dissolved oxygen in their immediate environment.
Reason (R) : Algae are the primary producers of energy rich compounds which form the basis of food cycles of all aquatic animals.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

117. In gymnosperms, transfer of pollen grains is carried out by
(1) Insects (2) Birds
(3) Water current (4) Air current

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118. Read the following statements and identify the characters related to the alga shown in the diagram



(a) It is a member of Phaeophyceae
(b) Food is stored in the form of starch
(c) Rich in iodine
(d) Edible
(e) It shows dominance of fucoxanthin
Choose the **correct** answer from the options given below:
(1) (b), (c) and (e) only (2) (a) and (c) only
(3) (b), (c) and (d) only (4) (a), (c), (d) and (e) only

119. Sex organs of bryophytes are
(1) Multicellular and without jacket
(2) Located on gametophyte
(3) Located on sporophyte
(4) Always diploid

120. Select the **incorrectly** matched pair.
(1) *Ectocarpus* – Filamentous and branched
(2) *Chlamydomonas* – Pyrenoids contain protein besides starch
(3) *Sargassum* – Sexual reproduction takes place by fusion of non-motile gametes.
(4) *Polysiphonia*- The stored food is very similar to amylopectin and glycogen in structure

121. Which of the following statement is **incorrect** regarding liverworts?
(1) The female sex organ is called archegonium
(2) The male gametes are non-motile
(3) Gemmae are green and multicellular in *Marchantia*
(4) Liverworts grow usually in moist and shady habitats

122. What kind of division will occur in zygote and spore mother cell respectively in bryophyte?
(1) Mitosis & Mitosis (2) Mitosis & Meiosis
(3) Meiosis & Mitosis (4) Meiosis & Meiosis

123. Match the column I with column II.

	Column I		Column II
a.	*Gracilaria*	(i)	Colonial form with oogamous type of reproduction
b.	*Volvox*	(ii)	Homosporous with distinct strobili
c.	*Funaria*	(iii)	Sexually reproduces by non-motile gametes
d.	*Equisetum*	(iv)	Shows an elaborate mechanism of spore dispersal

Choose the **correct** option.
(1) a(i), b(iii), c(iv), d(ii) (2) a(iii), b(i), c(ii), d(iv)
(3) a(iii), b(i), c(iv), d(ii) (4) a(iv), b(i), c(iii), d(ii)

124. In pteridophytes, the embryo formed after the fusion of gametes
(1) Develops into seed
(2) Develops into a multicellular well differentiated sporophyte
(3) Gives rise to female haploid structure
(4) Forms unicellular structure

125. In gymnosperms, nucellus is protected by envelope and this composite structure is called
(1) Endosperm (2) Integument
(3) Ovule (4) Ovary

126. Which of the following is **not** a member of gymnosperm?
(1) *Pinus* (2) *Ginkgo*
(3) *Cycas* (4) *Adiantum*

127. Identify the statement which is **not** true.
(1) Plant body of algae is less differentiated than that of bryophytes
(2) Mosses lack true roots, stems and leaves
(3) Gametophyte of pteridophytes require cool and damp places to grow
(4) Gymnosperms are homosporous and produce haploid spores

128. One of the tallest gymnosperms is
(1) *Sequoia* (2) *Pinus*
(3) *Ginkgo* (4) *Cycas*

129. Strobili or cones are not found in
(1) *Selaginella* (2) *Equisetum*
(3) *Pinus* (4) *Polytrichum*

130. *Cycas* lacks
(1) Coralloid roots (2) Female cone
(3) Pollen grains (4) Unbranched stem

131. Archegonia is absent in
(1) *Polytrichum* (2) *Pinus*
(3) *Eudorina* (4) *Pteris*

132. Leaves of conifers do **not** have/show
(1) Thick cuticle (2) Sunken stomata
(3) Reduced surface area (4) High water loss

133. Read the following statements and select the **correct** option.
Statement A: In angiosperms, parenchyma tissue forms the major component within organs.
Statement B: Parenchymatous cells have the cell wall composed of cellulose and lignin
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

134. Pteridophytes differ from gymnosperms in
(1) Being tracheophytes
(2) Having sporophyte as main dominant body
(3) Having independent gametophytic bodies
(4) Being archegoniates

135. Which of the following features is shared by the three main classes of algae viz Chlorophyceae, Phaeophyceae and Rhodophyceae?
(1) Vegetative mode of reproduction
(2) Flagellar number and position of insertions
(3) Type of stored food
(4) Cell wall composition

ZOOLOGY

136. Which of the following substances is not excreted through sweat?
(1) NaCl (2) Urea
(3) Waxes (4) Lactic acid

137. Despite high blood pressure in glomerular capillaries, plasma proteins normally do not appear in the filtrate because
(1) Filtration is selective for only waste molecules
(2) They are degraded before filtration
(3) They are too large to pass through the capillary pores
(4) They are filtered actively

138. Human kidneys can produce urine nearly ____ times concentrated than the initial filtrate formed.
Choose the correct option to fill in the blank.
(1) Two (2) Three
(3) Four (4) Five

139. Which of the following structures allows passage of small amounts of urea into the medullary interstitium to maintain the osmolarity?
(1) DCT (2) Collecting duct
(3) PCT (4) Bowman's capsule

140. What fraction of cardiac output is filtered by kidneys per minute in humans?
(1) 1/2 (2) 1/3
(3) 1/5 (4) 1/4

141. Extensions of cortex between medullary pyramids of kidney are known as
(1) Columns of Bertini (2) Ducts of Bellini
(3) Major calyces (4) Renal papilla

142. During the process of tubular reabsorption in humans, all of the following substances are reabsorbed actively, **except**
(1) Glucose (2) Amino acids
(3) Na^+ (4) Urea

143. About _____ of urea is excreted out per day by an adult human. Choose the option which fills the blank **correctly**.
(1) 5-10 mg (2) 25-30 mg
(3) 25-30 g (4) 2.5-3 mg

144. Reabsorption of water does **not** occur in human nephron in
(1) Collecting duct
(2) Proximal convoluted tubule
(3) Distal convoluted tubule
(4) Ascending limb of loop of Henle

145. Which of the following organism is ammonotelic?
(1) Mammals (2) Reptiles
(3) Land Snails (4) Most bony fishes

146. In the nephron, the maximum amount of water is reabsorbed in the
(1) Proximal convoluted tubule
(2) Loop of Henle
(3) Distal convoluted tubule
(4) Collecting duct

147. Which of the following is added into the blood drained from an artery before pumping it into the dialysing unit?
(1) Serotonin (2) Histamine
(3) Heparin (4) Platelets

148. The walls of the atria of the heart release ANF in response to
(1) Increase in blood volume and pressure
(2) Decrease in blood volume and pressure
(3) Increase in blood volume but decrease in blood pressure
(4) Decrease in blood volume only

149. Path of urine drainage in the human excretory system is
(1) Calyx → Collecting duct → Renal pelvis → Ureter → Urinary bladder
(2) Renal pelvis → Calyx → Collecting duct → Ureter → Urinary bladder
(3) Collecting duct → Calyx → Renal pelvis → Ureter → Urinary bladder
(4) Renal pelvis → Collecting duct → Calyx → Ureter → Urinary bladder

150. Protonephridia or flame cells are the excretory structures present in
(1) Prawns, cockroaches (2) Earthworm, *Ascaris*
(3) Rotifers, earthworm (4) *Planaria*, *Amphioxus*

151. In descending limb of loop of Henle, filtrate becomes
(1) Hypotonic to plasma (2) Isotonic to plasma
(3) Hypertonic to plasma (4) Isotonic to lymph

152. Kidneys are located close to the dorsal inner wall of the abdominal cavity between the levels of
(1) 12th thoracic and 3rd lumbar vertebra
(2) 7th thoracic and 12th thoracic vertebra
(3) Last sacral and 1st caudal vertebra
(4) 1st lumbar and 5th lumbar vertebra

153. The approximate amount of CO_2 removed by the lungs of a healthy adult human in an hour is
(1) 200 mL (2) 6000 mL
(3) 12000 mL (4) 8000 mL

154. Hormone produced by the kidneys is
(1) Aldosterone (2) Angiotensinogen
(3) Rennin (4) Erythropoietin

155. A person is experiencing dehydration due to diminished ability of kidneys to conserve water leading to excessive water loss. Deficiency of which hormone and damage to its source gland respectively could possibly cause this?
(1) Vasopressin : Adenohypophysis
(2) ACTH : Neurohypophysis
(3) ACTH : Adrenal cortex
(4) ADH : Hypothalamus

156. According to sliding filament theory, which of the following slides over the thick filaments to shorten the muscle during its contraction?
(1) Myosin filaments (2) Meromyosins
(3) Thin filaments (4) Collagen fibres

157. Choose the correct option to complete the analogy for the following
Flagellar movement : Spermatozoa : : Amoeboid movement : _____
(1) Ciliated epithelial cells (2) RBCs
(3) Leucocytes (4) Cuboidal epithelial cells

158. Read the statement w.r.t muscles and choose the option that correctly fills the blank.
In the resting state, a subunit of _____ masks the active binding sites for myosin on the actin filaments.
(1) Tropomyosin (2) F actin
(3) G actin (4) Troponin

159. During muscle contraction, the length of thin filaments
(1) First increases then decreases
(2) Remains unchanged
(3) Increases
(4) Decreases

160. Which of the following is primarily responsible for transmitting impulses from a motor neuron to a skeletal muscle fibre?
(1) Norepinephrine (2) Serotonin
(3) Acetylcholine (4) GLUT-4

161. Contractile protein 'X' generates force during muscle contraction along with the protein 'Y', whereas 'Z' is a structural protein (absence of which is seen in muscular dystrophy).
Identify the proteins and choose the most appropriate option for 'X' and 'Z' respectively.
(1) F-actin and troponin (2) Troponin and G-actin
(3) Myosin and dystrophin (4) Tropomyosin and actin

162. Which of the following is a storehouse of calcium in a muscle fibre?
(1) Cytosol (2) Sarcoplasmic reticulum
(3) Myofilaments (4) Sarcolemma

163. Choose the **incorrect** statement w.r.t. types of movement in the human body.
(1) Contractility and excitability are common properties exhibited by both neural and muscular fibres
(2) Ciliary movement occurs in hollow tubular organs
(3) Macrophages and leucocytes exhibit amoeboid movement
(4) Passage of dust particles in respiratory tract is facilitated by ciliary movement

164. The type of muscle fibres present in our thighs are
(1) Striated and involuntary
(2) Unstriated and involuntary
(3) Unstriated and voluntary
(4) Striated and voluntary

165. Cytoskeletal elements like microfilaments are exclusively involved in which type of movement?
(1) Amoeboid (2) Ciliary
(3) Muscular (4) Flagellar

166. Read the following:
(a) Globular protein
(b) Trimeric
(c) Regulatory protein
All of the above are true for
(1) Tropomyosin (2) Troponin
(3) G-actin (4) Meromyosin

167. A large number of myofibrils in sarcoplasm of skeletal muscle fibres are arranged _____ w.r.t. the long axis of the muscle fibre, showing alternate dark and light bands.
Choose the option which fills the blank **correctly**
(1) Radially (2) Parallely
(3) Horizontally (4) Obliquely

168. During heavy exercise, muscles undergo fatigue and contain
(1) More ATP, less glycogen
(2) More ADP, more lactic acid
(3) More creatine phosphate, less lactic acid
(4) More creatine phosphate, more ATP

169. Smooth muscle fibres differ from skeletal muscle fibres in all of the following, **except**
(1) Absence of striations
(2) Location
(3) Presence of actin and myosin
(4) Number of nuclei in a cell

170. The active ATPase enzyme and binding sites for ATP are present on
(1) Short arm of meromyosin
(2) Light meromyosin
(3) Cross arm of light meromyosin
(4) Globular head of meromyosin

171. A vertebrate skeleton can be divided into two sub-skeletons. The 'A' skeleton includes skull and vertebral column and the 'B' skeleton is made up of limb bones and their girdles.
Choose the option that fills the blanks **correctly**

	A	B
(1)	Exo	Endo
(2)	Appendicular	Axial
(3)	Axial	Appendicular
(4)	Endo	Exo

(1) (1) (2) (2)
(3) (3) (4) (4)

172. The type of synovial joint which is present between humerus and pectoral girdle is also present between
(1) Atlas and axis
(2) Femur and tibia
(3) Carpal and metacarpal of thumb
(4) Femur and pelvic girdle

173. In adult humans, the total number of phalanges present in each limb is equal to the total number of
(1) False ribs (2) Tarsals in upper limb
(3) True ribs (4) Lumbar vertebrae

174. Lower jaw in humans is made up of
(1) Mandible (2) Vomer
(3) Maxilla (4) Palatine

175. Choose the **odd** one w.r.t. joints associated with cervical vertebrae in humans.
(1) Synovial joint (2) Pivot joint
(3) Fibrous joint (4) Cartilaginous joint

176. The long slender bone which is a part of pectoral girdle and characterised by two curvatures, articulates with
(1) Coxal bone (2) Acromion
(3) Acetabulum (4) Glenoid cavity

177. In humans, at the point of fusion of ilium, ischium and pubis, there is a cavity to which the thigh bone articulates. This cavity is called
(1) Glenoid cavity (2) Acromion
(3) Acetabulum (4) Pubic symphysis

178. In an accident, a man suffered from a severe fracture in the longest bone of his body. This longest bone is called
(1) Tibia (2) Sternum
(3) Femur (4) Patella

179. All true ribs are A attached to the thoracic vertebrae and B connected to the sternum with the help of hyaline cartilage.
Choose the option that **correctly** fill the blanks A and B respectively.
(1) Ventrally, Medially (2) Dorsally, Ventrally
(3) Laterally, Ventrally (4) Laterally, Dorsally

180. Each myofibril contains many serially arranged units called
(1) Muscle fibre (2) Sarcomere
(3) Myofilament (4) Muscle bundle